Multi-function air disinfection machine operation manual of

Y1000

Laoken brand with KDSJ-Y1200 model

Y1500

Manufactured by Laoken Medical Technology co. , LTD

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Ⅰ. Product model

KDSJ-Y□□

design code （A B C....）

Air circulation air volume （unit: m /h）

Mobile

Product code

EG: KDSJ-Y1000 the meaning is Laoken medical technology co., LTD product fit for 100M space, mobile multi-function air disinfection machine

Ⅱ. Product structure

This equipment is to use wind circulation principle of ultraviolet disinfection, and combined with filtering, anion and fresh air, realize sustainable disinfection of indoor air. Sterilization machine is mainly composed of filter, high intensity ultraviolet lamp, anion generator, fan, control system, the human body infrared sensor and chassis, etc.

1、air outlet

2、controller(display screen)

3、The human body infrared sensor

4、air inlet(initial effect/ medium-efficiency filter)

5、power line/protective tube

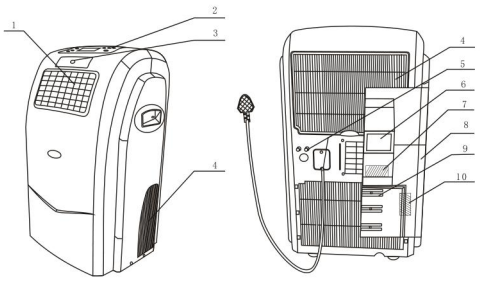
6、draught fan

7、electronic ballast

8、main case

9、ultraviolet lamp

10、negative-ion generator



Ⅲ. Product features and characteristics

1. Disinfection effect: White staphylococcus (8032) the killing rate of 99.90% or more; the death of natural bacteria in air rate of 90.00% or higher.

2. Filter dust removal aseptic: With physical method to filter dust and microorganism in

the air, air carried out in cooperation with ultraviolet disinfection. At the same time effectively prevent the dust on the influence of ultraviolet lamp radiation intensity. Disinfection machine filter for removable polymer antistatic filter sponge.

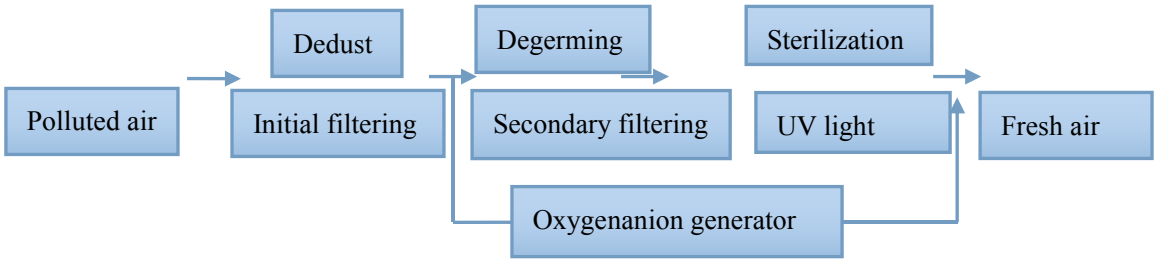
3. High intensity ultraviolet disinfection: using ultraviolet disinfection lamp and science combined into a radiation intensity of more than 10000 mu W/cm high intensity ultraviolet (uv) radiation zone, under the action of indoor air in fan cycle by ultraviolet radiation zone and disinfection treatment.

4. Negative oxygen ion purify air: Air negative oxygen ion is known as "air vitamin", is the need of human life and health, a certain amount of negative oxygen ions, will improve the indoor air quality, and is advantageous to the staff in good health.

5. Safe and reliable: disinfection machine adopts physical sterilization method, during the working process of the machine, and no harmful gas and material generated. The high intensity UV light was enclosed in the air duct, and leakage is less than 1 1μW/cm, O_3 concentration is less than 0.10mg/m. No harm to people and equipment.

6. Easy to use: manual disinfection, automatic disinfection by infrared ray, program-controlled automatic disinfection, infrared remote control operation regularly.

Disinfection principle



IV. Main parameter

Model	KDSJ-Y1000	KDSJ-Y1200	KDSJ-Y1500
Air circulation volume（m^3/h）	≥1000	≥1200	≥1500
Applicable area（m^3）	100	120	150
UV radiation intensity inside（μw/ cm^2）	≥10000		
UV light leakage outside（μw/ cm^2）	≤1		
Working life of UV lamp（h）	≥5000		
O_3content in process（mg/ m^3）	≤0.1		
Input power（W）	220±10%		

Noise（dB）	≤50
Fuse	F2.5AL250V
Net weight（Kg）	24
Gross weight（Kg）	27
Unit dimension（L*W*H）mm	510*420*850
Package L*W*H）mm	610*486*940

Working condition: temperature range:5 ℃ ～ 40 ℃ ≤80 % Power: 220V±22V 50Hz±1Hz Humidity:

Sterilization lasting time:2～3h

Transportation and storage:

Environment temperature range:－20℃～50℃; Humidity≤90%; Atmospheric pressure range:86kPa～106kPa.

V. Application

This product is widely used in operating room, medical room, wards.

VI. Symbols

Notice! Check attached files	High voltage Cut power supply firstly, and read user manual.	Cut electricity (general power)	Connect electricity (general power)	B model applied section

Attention:

● Ignoring regulations will cause users or maintenance person injuries. The device adopt circulate air ultraviolet disinfection principle, and already done UV leakage measures, and no need for electromagnetic interference and protective measures.

● The user manual should be kept carefully.

● Please contact us if any problem in the user manual.

● Please read carefully the user manual before use and maintenance.

VII. Attention

1. Close window and doors whether used for manual disinfection or automatic

disinfection.

2. No articles covering or sheltering air intake and outtake of the equipment.

3. Socket of power supply must be 3 core socket with safety ground wire.

4. Water proof is seriously needed and power supply must be cut off when using wet fabric cleaning the equipment.

5. Check operation condition of the equipment regularly. Any abnormal should be examined and repaired in time. Electric failure should be done by professional operators. Please take out 7# battery when the remote will not be used for a long time.

VIII. Installation

Choose proper installation locality, place disinfecting equipment and it would work as soon as plug in. the best place is in the middle of one wall. Please assure no articles covering or sheltering air intake and outtake of the equipment. Meanwhile human infrared sensing device should be aligned people, or start/stop function can not be operated normally.

IX. Operation method

Please power on firstly.

If lights, motors, negative ion damage, "Error tips" and "lights, motors and negative ion" on display start flashing. If working time over setting range of UV lamp and filter, "UV lamp" and "Filter" on display normal on.

1. Manual disinfection

In standby mode, press "Start / Stop" button, start 1 hour disinfection; display "Remaining time 01:00"; In standby mode, press 1 to 9, then press "Start / Stop" to change disinfection hours. Press "start / stop" during cycle will back to standby mode.

2. Automatic disinfection

In standby mode, press "Automatic" button once, if the machine detected continuously human activity about 5 seconds, it will start disinfection and show "Automatic disinfection 00:30"; if without human activity is detected in 30 minutes, the disinfection will be turned off; press the "Automatic" button once, automatic function will be turned off, "Automatic" words extinguished.

3. Program control disinfection

In standby mode, press "Timing" button, display "Program control". At this time if the current time and the boot time consistent, air disinfection machine automatically enter working state; Press "Timing" again, programmed stop disinfection, back to the standby state. In this case, the startup and shutdown time is invalid.

4. Set the clock

In standby mode, press "Time" key once, left side 2 digits flashing. At this time press the number keys set hours (for 24 hours), press "Time" key again, right side digital flashing. At this time press the number keys to set minutes, again press "Time" key to save the time setting; in the setting if no operation in 5 seconds, system will exit to standby mode, unfinished setting will be canceled. After setting the clock, restart the machine, the setting remains.

5. Automatic on / off time setting

In standby mode, press "Program" button, the screen display "Boot time group 1", then press the number keys to set the boot time (see setting clock); then click "Program" button, the screen display "Shut down time group 1", press the number keys set off (see time clock); then click" program "button, the screen display " Boot time group 2 ", then press the number keys to set the boot time (see setting clock); then click" program "button, the screen display "Shut down time group 2", according to the timing shutdown time the number keys set off time (see setting clock); two groups should not cross, otherwise there is no interval between two groups.

6. Show cumulative working hours

In standby mode, press "Mode" key, the display shows the cumulative working hours.

7. Show UV intensity

When the cycle finished, press the number"0" key once, the screen displays the current state of the intensity of the UV light, such as the current intensity of 10000μW, showing "10000μW". After 10 seconds it will automatic return to the working interface.

8. Reset：

Press “Mode” one time when machine standby, and when display shows working time, press “0” button 5 times to enter “ Reset”.

a. Light working time: In the condition of cumulative working hours, press the "mode" key to switch to the bulb use time display, First display "L" 1 second then show lamp current use time "xxxxxH". Number keys can be changed, changed and save. Set to "00000 h" reset

b. Filter working time: Continue to press the "mode" key switch filter using time, according to a first display 1 second "E" and then "xxxxxH"

c. Press "stop" button to exit the Settings, or wait for 5 seconds to cancel Settings.

10. Maintenance

1. Filter maintenance：Depending on the work environment, when the machine filter dust too much, it should be timely clean or replace filter. When changing, open the machine mask, taken from the machine filter, clean with clear water filter or change into a new, finally it back into the machine.（Half a year）

2. The maintenance of the UV lamp：Ultraviolet radiation intensity of high and low will affect the air disinfection effect, therefore, regularly used alcohol cotton ball or

gauze wipe UV lamp and reflector, to ensure the radiation intensity of ultraviolet lamp is not affected by the dust.

3. Negative oxygen ion generator maintenance：Due to the effect of the dust of the negative oxygen ion, use for a long time, after the generator outlet near the casing will deposit a lot of dust, therefore, should be cleared on a regular basis. First of all, to cut off the power when clean, wipe with a soft dry cloth or a little alcohol. Note: cannot wash with water, the human body induction sensor and display cannot use any detergent, alcohol, etc. to wipe, can only use a wet towel to wipe gently.

This equipment and accessories at the end of its life will cause the risk：

1. UV lamp, electronic ballast, influence the air disinfection effect；
2. Filter net：Influence the air dust filtering effect；
3. Battery：It will affect the accuracy of the machine function of remote control and Beijing time

11. Troubleshooting

No.	Malfunction	Reasons	Resolutions
1	Machine not working	Power supply without power or power plug contact undesirable	Check the power supply and power outlet or the power cord
		Power supply fuse broken	Security after power, replace the insurance tube
2	Display show light with alarm	In addition to the standby lamp, lamp doesn't work（Ballast is damaged, the lamp and lamp holder poor contact, tube damage）	replace the ballast, check the lamp holder, replacement bulbs
		Ultraviolet radiation intensity is lower than the specified value（Light with dust, lamp aging attenuation, power supply voltage is too low, can't start the spare tubes）	Cleaning, replacement, measure the supply voltage, arranged check spare light.
3	The air is fresh effect is poor	Negative oxygen ion generator fault	Negative oxygen ion generator
		Filter, electrostatic dust collection box	Cleaning filter and the collection box
		Activated carbon membrane filter failure	Regeneration after use
4	Program control cannot start to work or stop the device,,	Program control off	Press “timing” button and make sure “timing” on
		Program controlled on/off time is not set	Please set the correct on/off time
		Time setting inaccurate	Set the local time correctly

	cannot start the human infrared induction automatically monitor disinfection	Human body sensor is not well plugged	Check human body sensor
		Infrared induction automatically disinfection is off	Press “Auto” key to let “Auto” displays in screen
		Human body induction sensor does not target the inspection zone	Adjust the induction sensor to target the inspection zone
5	Device cannot start to work with alarm	Fan cannot work	Check fan or check whether cable connects well
6	Fan cannot start to work or fan speed too slow	Start capacity is broken, or too low voltage, or connector is not well connected	Replace capacity, check voltage, check the connector
7	Too little wind in wind outlet	Too much dust in filter	Replace filter

Caution：

1. Replace fuse, please make sure that the device is already shut off, then use screw driver to open fuse socket to take the broken fuse out, then put the correct fuse into the socket and close it.
2. When replace the UV lamp, please make sure that the device is already shut off, then cut the bundle to take the UV lamp out, then assemble the correct lamp.
3. When replace the electronic ballast, please make sure that the device is already shut off, then use screw driver to loosen the wire of electronic ballast, then use M4 sleeve to remove the screw which is used to fix the electronic ballast, then replace it and fix it well.
4. Only the professional maintenance staff or personnel who is authorized by our company can check and repair our device. Please don’t open our device to check without permission.

Xii Trouble shooting

Dear user, from the day you buy our device, our good after sales service will follow you to solve all device’s problem.

● Principles of warranty

1. The device has one year warranty.
2. Any distributor ask you to pay cost during warranty period is forbidden.

● Whole life span repair

Our company will repair our device during its life span, you will just pay the repair cost.

● Free guidance

Our engineer will freely answer all your technical questions, guide you how to operate, and also will give you training to make you run our device perfectly.

● Other caution

If you buy our device from any distributor, also you can contact us to have good after sales service.

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